

RAJ MAHESHWARI

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Education

Carnegie Mellon University - School of Computer Science

Aug 2025 – Dec 2026

Master's in Natural Language Processing and Machine Learning (MIIS, LTI); GPA: 3.96/4.0

Pittsburgh, PA

- **Selected Coursework:** Advanced NLP, Deep Learning Systems, Generative AI, LLM Systems, LLM Applications.
- **Research** (advised by **Prof. Yonatan Bisk**; under review at **EMNLP 2026**): Evaluating data augmentation using consistent and counterfactual examples to improve contextual faithfulness while preserving parametric knowledge across SFT and DPO.
- **Graduate Teaching Assistant:** Probability & Computing, Generative AI.

International Institute of Information Technology, Hyderabad

Aug 2019 – Jul 2023

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.54/10

Hyderabad, India

- **Coursework:** Data Structures & Algorithms, Linear Algebra, Probability, Operating Systems & Networks, Computer Vision, Topics in Reinforcement Learning, Distributed Systems, Data & Applications.

Experience

Machine Learning Research Intern

May 2026 – Aug 2026

OpenHands | advised by Prof. Graham Neubig

Pittsburgh, PA

- Architected an **8-stage end-to-end pipeline** turning **200K+** real OpenHands agent conversations (hundreds of GB of OpenTelemetry traces across **1,000+** unique repositories) into verifiable, leak-proof RL coding tasks in **Harbor** format. Automated creation of fail-to-pass and pass-to-pass tests for task grading using a **multi-agent framework**.
- Enabled a **recursively self-improving mechanism** for the **OpenHands SDK**, using meta-harness optimization and verifier rewards to improve agent task performance within the resulting RL environment.

Software Engineer

Jul 2023 – Jul 2025

Google – Search Verticals

Bengaluru, India

- Integrated sports insights into **Samsung S25 Now Bar**. Enriched LLM-generated outputs with real-time first- and third-party data streams to deliver fresh, glanceable, and context-rich updates for hundreds of millions of sports fans worldwide.
- Developed sports experiences across multiple surfaces – Discover, Home Stack, iOS and Android apps, and notifications. Contributed to **Sports Hub**, a personalized web app delivering real-time updates and content for fans.
- Improved engagement in the **Google Sports MiniApp** by enabling users to discover and follow teams & leagues through new search functionality. Designed backend pipelines with Knowledge Graph queries to maintain active datasets across **50+** locales.
- Automated migration of **1,500+** server platform nodes to a new integration-testing framework, reducing manual effort, improving onboarding efficiency, and **saving an estimated several decades of cumulative engineering effort**.
- Recognized as a coach for the **Google Tech Immersion** program, mentoring and onboarding Nooglers, fostering confidence, productivity, and a stronger engineering culture across teams.

Publications

Mind the Gap: Multilingual Divide in LLM Bias Detection and Reasoning.

ACL 2026 SRW

Public Blockchain-Envisioned Security Scheme for Internet of Drones Applications.

IEEE TVT 2023

Projects

Agentic Concierge | *Python, FastAPI, SQLite*

Mar 2026 – Apr 2026

- Built a **harness for agentic booking** across rides and appointments, using **pluggable adapters** for providers and domains.
- Enforced safe, auditable execution with a **propose-then-commit gate** and append-only, SHA-256 hash-chained logs, preventing replay and parameter substitution while enabling session reconstruction and tamper detection.

TaskWeaver | *PyTorch, Hugging Face*

Aug 2025 – Nov 2025

- Developed a **hypernetwork-based system** for instance-level model adaptation that generates **per-input LoRA weights** from prompt embeddings, extending **Text-to-LoRA**.
- Learned a **task-label-free** universal adaptation mapping across Pythia, Gemma, and Qwen, outperforming mixed-LoRA on held-out **RACE** (18.7% → 25.1%) and **SVAMP** (4.67% → 16.3%).

Summer of Bitcoin | *JavaScript, Node.js*

Jul 2022 – Sep 2022

- **Open Source Contributor** for a **Bitcoin node/wallet system**. Implemented **key feature upgrades** to improve scalability, reliability, and test coverage. Mentored contributors on blockchain and cryptocurrency fundamentals.

Skills

Languages: Python, C/C++, Java, Go, JavaScript, Bash

Frameworks & Libraries: PyTorch, NumPy, Pandas, React, Flutter

Achievements

AIR 1149, IIT JEE Advanced (top 0.5%); **AIR 1160, IIT JEE Main** (top 0.1%)

2019